EMBEDDED SOFTWARE ENGINEER

LAITH ARMOUTI

CONTACT INFORMATION

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https://github.com/laithalarmouti

SKILLS

- Embedded Software Development
- Firmware Development
- Embedded C
- C++
- ARM Cortex-M
- Bare-Metal Programming
- Real-Time Systems
- STM32
- FreeRTOS
- Device Drivers
- Communication Protocols • I2C • SPI • UART • CAN
- TensorFlow Lite for Microcontrollers
- CMSIS-DSP
- Sensor Data Processing
- On-device Inference
- STM32CubeIDE
- Git
- GitHub
- MS Powerpoint
- MS Word
- MS Excel

LANGUAGES

- English | Native
- German | Intermediate
- Arabic | Native

SUMMARY

Embedded Software Developer with strong experience in firmware and embedded systems development for ARM Cortex-M microcontrollers, specializing in STM32 platforms, Embedded C/C++, and real-time systems. Hands-on experience with bare-metal firmware, FreeRTOS, interrupt handling, and low-level peripheral drivers (GPIO, I2C, SPI, UART, CAN). Proven ability to deploy TensorFlow Lite for Microcontrollers (TFLite Micro) for on-device AI inference, apply CMSIS-DSP for sensor signal processing, and optimize performance in memory-constrained embedded systems. Experienced in debugging, hardware bring-up, and developing reliable embedded software from prototype to deployment.

EXPERIENCE

JUNIOR EMBEDDED SOFTWARE DEVELOPER

Kalamntina. Amman, Jordan | Jun 2022 - Jun 2023

- Designed and developed embedded software solutions, contributing to the optimization and improvement of existing embedded platforms.
- Implemented device drivers and software libraries in C/C++ for STM32 microcontrollers and other MCU-based systems.
- Applied RTOS concepts, including task scheduling, synchronization, testing, and verification, to support real-time embedded applications.

WEB DEVELOPER

Alameed Coffee. Amman, Jordan | Jun 2023 - Jan 2024

- Developed full-stack web applications using the MERN stack (MongoDB, Express, React, Node.js) to support business operations and improve user experience.

JUNIOR SOFTWARE DEVELOPER

Kaizen Plus. Amman, Jordan | Jan 2024 - May 2024

- Developed a web-based employee tracking and risk-avoidance system for ADNOC (UAE) using .NET Core, Angular, and Redis, enabling real-time monitoring of employee locations and improving operational safety.

BACK-END DEVELOPER

Trinitae Duftkultur GmbH. Bonn, Germany | May 2024 - Jul 2024

- Supported the launch of Trinitae Duftkultur's official website

EDUCATION

BACHELOR OF SCIENE IN COMPUTER ENGINEERING

Girne American University. Cyprus | Jul 2017 - Jul 2021

MASTER OF ENGINEERING IN COMPUTER SCIENCE

Gisma University Of Applied Sciences . Potsdam, Germany | Jul 2024 - Jul 2025

- Specialized in embedded systems, with a focus on AI-powered microcontroller applications, including research on fall detection using STM32 and TensorFlow Lite.

PROJECTS

Embedded Fall Detection System (STM32)

- Designed and developed an embedded real-time system on the STM32F411RE (ARM Cortex-M4) for motion-based fall detection using MPU6050 IMU sensors.
- Implemented low-level sensor interfacing and data acquisition, including I2C communication, interrupt handling, and buffered sampling on the microcontroller.
- Applied windowing logic and CMSIS-DSP routines for embedded signal processing and feature extraction under memory and timing constraints.
- Integrated a lightweight TensorFlow Lite for Microcontrollers model into the firmware to perform on-device classification, optimized for low latency and limited RAM/Flash.
- Measured and validated real-time performance using SysTick, ensuring deterministic execution suitable for embedded and wearable systems.
- Evaluated single-sensor vs. dual-sensor configurations, balancing system complexity, resource usage, and detection performance for practical embedded deployment.

Additional Embedded (Hobbyist) Projects

- Built personal embedded projects using available hardware to explore microcontroller programming and sensor integration.
- Integrated temperature sensors, IMU sensors, and other peripherals using GPIO, I2C, SPI, and UART.
- Focused on embedded C/C++, firmware development, and basic real-time behavior on resource-constrained systems.
- Published project source code on GitHub.

CERTIFICATIONS & COURSES

Mastering Microcontroller and Embedded Driver Development FastBit Embedded Brain Academy

- Embedded C, bare-metal programming, ARM Cortex-M
- Memory-mapped registers, low-level driver development
- GPIO, UART, I2C, SPI peripheral drivers

Mastering Microcontroller: Timers, PWM, CAN, Low Power FastBit Embedded Brain Academy

- Timers, PWM, CAN bus
- Real-time communication
- Low-power embedded system design

Mastering RTOS: Hands-On FreeRTOS and STM32Fx with Debugging FastBit Embedded Brain Academy

- FreeRTOS, task scheduling
- Queues, semaphores, mutexes
- Interrupt handling, real-time debugging